ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications [ANSWER KEY]

Unit 5.01: Topic 5: Differential & Integral Calculus

Q1. Consider the function fx = x^2 - 5x + 6 . What is the value of f2 ?

[x] A: 0
[ ] B: -2
[ ] C: 4
[ ] D: 8

Solution: To find the value of f2 , substitute x=2 into the function fx = x^2 - 5x + 6 . f2 = 2^2 - 52 + 6 f2 = 4 - 10 + 6 f2 = 0 [M1] [A1]

Q2. Consider the function fx = x^2 - 4x + 3 . The graph of this function is shown below. Find the y-intercept of the graph.

[x] A: 3
[ ] B: -1
[ ] C: 0
[ ] D: -3

Solution: The y-intercept of a function is the value of the function when x=0. For the function fx = x^2 - 4x + 3 , we can find the y-intercept by substituting x=0: f0 = 0^2 - 40 + 3 f0 = 0 - 0 + 3 f0 = 3 Thus, the y-intercept is 3. [M1] [A1]

Q3. Consider the function fx = x^2 - 2x + 1 . Determine the y-coordinate of the vertex of the graph of y = fx .

[ ] A: 1
[x] B: 0
[ ] C: -1
[ ] D: 2

Solution: The graph of a quadratic function fx = ax^2 + bx + c is a parabola. The x-coordinate of the vertex is given by x = -frac{b}{2a} . For the function fx = x^2 - 2x + 1 , we have a=1 and b=-2 . Therefore, the x-coordinate of the vertex is x = -frac{-2}{21} = frac{2}{2} = 1 . To find the y-coordinate of the vertex, we substitute this value of x back into the function: f1 = 1^2 - 21 + 1 = 1 - 2 + 1 = 0 . Thus, the y-coordinate of the vertex is 0. [M1] [A1]

Q4. Consider the function fx = 3x^2 - 2x + 5 . Find the value of f'1 .

[x] A: 4
[ ] B: 5
[ ] C: 6
[ ] D: 7

Solution: The derivative of a function fx gives the instantaneous rate of change of the function at any given point. We first find the derivative of fx = 3x^2 - 2x + 5 using the power rule: f'x = frac{d}{dx}3x^2 - frac{d}{dx}2x + frac{d}{dx}5 . Applying the power rule, frac{d}{dx}ax^n = nax^{n-1} and the derivative of a constant is 0, we get f'x = 23x^{2-1} - 12x^{1-1} + 0 , which simplifies to f'x = 6x - 2 . To find the value of the derivative at x=1 , we substitute x=1 into the expression for f'x : f'1 = 61 - 2 = 6 - 2 = 4 . [M1] [A1]

Q5. The graph of the function fx = 2x^2 - 3x + 5 is shown below. Determine the value of f'2.

[ ] A: 7
[x] B: 5
[ ] C: 9
[ ] D: 2

Solution: The derivative of the function fx = 2x^2 - 3x + 5 is found using the power rule: f'x = frac{d}{dx}2x^2 - frac{d}{dx}3x + frac{d}{dx}5. This gives f'x = 4x - 3. To find f'2, substitute x=2 into the derivative: f'2 = 42 - 3 = 8 - 3 = 5. [M1] [A1]

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Q6. Consider the function fx = x^2 - 6x + 5 . Find the x-coordinate of the vertex of the parabola represented by y = fx .

[ ] A: 1

[x] B: 3

[ ] C: -1

[ ] D: -3

Solution: The given function is a quadratic function in the form fx = ax^2 + bx + c , where a=1 , b=-6 , and c=5 . The x-coordinate of the vertex of a parabola is given by the formula x = -frac{b}{2a} . Substituting the values of a and b, we get x = -frac{-6}{21} = frac{6}{2} = 3 . Therefore, the x-coordinate of the vertex is 3.
[M1] [A1]

Q7. Find displaystyleint x^{2}+4x^{3},dx .

[x] A: dfrac{1}{3}x^{3}+x^{4}+C

[ ] B: dfrac{1}{4}x^{4}+dfrac{4}{5}x^{5}+C

[ ] C: 3x^{3}+16x^{4}+C

[ ] D: x^{2}+4x^{3}+C

Solution: Working through this calculus step: Using the power rule int x^n,dx=dfrac{x^{n+1}}{n+1}+C : int x^{2}+4x^{3},dx=dfrac{1}{3}x^{3}+x^{4}+C .
[M1] [A1]

Q8. Find displaystyleint 5x^{2}+4x,dx .

[x] A: dfrac{5}{3}x^{3}+2x^{2}+C

[ ] B: dfrac{5}{4}x^{4}+dfrac{4}{3}x^{3}+C

[ ] C: 15x^{3}+8x^{2}+C

[ ] D: 5x^{2}+4x+C

Solution: Working through this calculus step: Using the power rule int x^n,dx=dfrac{x^{n+1}}{n+1}+C : int 5x^{2}+4x,dx=dfrac{5}{3}x^{3}+2x^{2}+C .
[M1] [A1]